ABAID ULLAH BHATTI

Software Engineer • Full-Stack Developer • Systems & Backend Lead

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EDUCATION

University of Engineering and Technology (UET), Lahore — **Lahore, Pakistan**

*Bachelor of Science in Computer Science — CGPA: **3.38 / 4.00*** — *2023 – 2027*

- **Honors & Awards:** Awarded Government Merit-Based Academic Scholarship (2020 – 2025).
- **Core Coursework:** Data Structures & Algorithms, Object-Oriented Programming, Database Systems, Operating Systems, Computer Networks, Software Engineering, Theory of Automata, Compiler Construction.

TECHNICAL SKILLS

Languages:	TypeScript, JavaScript (ES6+), Python, C, C++, SQL, x86 Assembly (MASM32), HTML5, CSS3
Frontend:	React 18, Next.js 16 (App Router), Tailwind CSS, Vite, Redux Toolkit, Component Architecture
Backend & APIs:	Node.js, Express.js, ASP.NET Core, RESTful APIs, JWT, Role-Based Access Control (RBAC), Middleware
Databases & Cloud:	PostgreSQL, MongoDB (Atlas/Mongoose), SQL Server, Redis, Supabase, Cloudinary, Docker
Tools & Testing:	Git, GitHub, Docker, Postman, Jest (58+ suites), Vitest, Supertest, Linux / Bash, CI/CD, Markdown
Engineering Focus:	AST Compilers, Database Benchmarking, Concurrency Transactions, Stripe Payment Integration

FLAGSHIP ENGINEERING PROJECTS

Litho: Markdown-to-LaTeX Publishing Ecosystem (Final Year Project) — **Python, AST Compiler, Tectonic TeX, Jinja2, CI/CD**

Compiler & LaTeX Backend Engine Lead — Department of Computer Science, UET Lahore — *2024 – Present*

- Architected an end-to-end technical publishing compiler mapping CommonMark AST nodes into versioned publisher LaTeX templates with lossless round-trip metadata recovery.
- Engineered a custom Jinja2 template-binding emitter generating high-fidelity LaTeX for callouts, math equations, citations, figures, and Gantt charts.
- Configured and packaged a trimmed Tectonic TeX engine fork with precompiled format files, accelerating PDF compilation times by over 40%.
- Built an automated golden-PDF visual regression testing harness in CI, verifying pixel-perfect diff rendering across IEEE, Springer, and thesis templates.

Shop.co — Enterprise MERN Stack E-Commerce Platform — **React 18, TypeScript, Node.js, Express, MongoDB, Stripe**

Full-Stack Architect & Developer — Live on Vercel & Render Cloud — *2024*

- Engineered a production-ready, fully-typed clothing commerce application with strict RBAC authorization and JWT authentication featuring access/refresh token rotation.
- Implemented unified shopping cart persistence supporting guest UUID sessions and automatic cart merging upon user authentication.
- Leveraged atomic MongoDB transactions to guarantee inventory consistency during concurrent multi-item checkouts and Stripe webhook fulfillment.
- Built an admin analytics dashboard with MongoDB native aggregation pipelines tracking revenue, product velocity, and customer cohort metrics.

Litho Template Marketplace — Curated Document Registry — **Next.js 16, React, Express 5, Supabase PostgreSQL, Docker**

Full-Stack Developer — Live on Render Cloud Docker Container — *2024*

- Developed a responsive template registry featuring dual-pane split view, real-time multi-token search (tag:, kind:), and a 10-theme CSS design system.
- Integrated Supabase PostgreSQL database with schema migration pipelines and built administrative moderation controls for submissions.
- Configured dual-deployment pipelines: static catalog export for GitHub Pages and production Docker web service on Render.

ENGINEERING PROJECTS (CONTINUED)

Cross-Database Performance Benchmarking Suite **Python, PostgreSQL, MongoDB, SQL Server, Redis, Matplotlib**
Solo Developer & Benchmarking Researcher *2024*

- Engineered an automated Python benchmarking framework measuring throughput and latency for CRUD, multi-table JOINs, and complex aggregation operations.
- Benchmarked SQL Server, PostgreSQL, MongoDB Atlas, and Redis under concurrent workloads, analyzing index optimization and query execution plans.
- Generated comparative visualization reports identifying performance sweet spots for relational vs. document vs. key-value storage paradigms.

Attendance Management System (AMS) **Next.js 14, React, Supabase PostgreSQL, Tailwind CSS**
Full-Stack Web Engineer *2024*

- Built an administrative attendance tracking platform with role-based access for faculty and administrators, daily attendance logging, and CSV exports.
- Engineered responsive interfaces with Tailwind CSS and integrated Supabase authentication with Row-Level Security (RLS) policies.

Secure Cyber Reconnaissance Framework **Python, Network Sockets, Information Security**
Lead Security Researcher *2024*

- Developed an early-warning reconnaissance framework mapping attack surfaces and identifying vulnerabilities in simulated enterprise networks using passive intelligence techniques.
- Reduced simulated threat vectors by over 30% through automated port auditing and misconfiguration discovery routines.

Graphical Chess Game in MASM32 Assembly **x86 Assembly, MASM32 SDK, Windows API**
Low-Level Systems Developer *2023*

- Implemented full chess rule verification and graphical board rendering using low-level Win32 GDI pixel manipulation and direct register allocation.
- Optimized memory footprint and CPU register usage to achieve zero-latency move validation in constrained 32-bit execution environments.

PROFESSIONAL EXPERIENCE & LEADERSHIP

Solo Full-Stack Engineer / Open-Source Contributor **Remote / Lahore, Pakistan**
Independent Software Development *2024 – Present*

- Architected and deployed 11 open-source software engineering repositories spanning full-stack web platforms, compiler toolchains, cybersecurity scanners, and low-level systems.
- Engineered production-grade security architectures including JWT authentication, rate limiting, helmet security headers, and comprehensive unit/integration test coverage with Jest.

UET ACM Student Chapter **UET Lahore, Pakistan**
Event Coordinator & Technical Volunteer *2023 – Present*

- Coordinated collegiate technical workshops, coding bootcamps, and hackathons for 300+ computer science students.
- Mentored junior undergraduates in introductory programming, version control workflows with Git, and data structures problem solving.

TECHNICAL CERTIFICATIONS & COURSEWORK

- **Full-Stack Web Development:** Comprehensive single-page application architectures, Next.js / React, and RESTful Express APIs (2024).
- **Relational Databases & SQL:** Advanced query optimization, schema normalization (1NF–BCNF), and ACID transactions (2024).
- **Python Programming & Scripting:** Object-oriented design, systems automation, and data pipeline scripting (2024).
- **MongoDB & Document Databases:** Aggregation pipelines, Atlas cluster administration, and Mongoose schema modeling (2024).
- **Modern JavaScript (ES6+) & Data Structures (DSA):** Asynchronous event loop, closures, and algorithmic problem solving (2023).